

Shamanthak Hegde

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SUMMARY

Machine Learning Engineer training and post-training LLMs and multimodal models up to 32B parameters on multi-node GPU clusters (DeepSpeed, FlashAttention, SLURM). Published at WACV and TMLR on multimodal models and preference optimization; TreeHacks 2026 Healthcare Track Winner for a real-time PHI detection system. Comfortable across the stack - SFT and RLHF/DPO/GRPO, high-throughput evaluation with vLLM, and shipping LLM agents and RAG backends on AWS with FastAPI.

EXPERIENCE

Graduate Research Assistant (Advisor: Dr. Yezhou Yang)

Sep 2023 – Present

Active Perception Group, Arizona State University

Tempe, AZ

- Worked on **Tau-Emotion-Bench**, a synthetic benchmark for evaluating LLM-based service agents under emotional context; built a **4-stage reverse generation pipeline** producing **280 evaluation tasks** and **1492 SFT trajectories** across airline, retail, and telecom domains.
- Trained **Qwen3-4B** and **Qwen3-8B** with SFT on the generated trajectories, achieving **+5.7pp** and **+3.9pp Avg@4 gains** on held-out τ^2 -bench tasks against GPT-4.1 and GPT-5.2 baselines while preserving baseline τ^2 -bench performance.
- Served fine-tuned Qwen3-4B/8B models** and frontier baselines on a multi-GPU server using **vLLM and SGLang** for Tau-Emotion-Bench evaluation, **configuring request batching** and **tensor parallelism** to enable **high-throughput Avg@4 rollouts** across 280 test tasks.
- Benchmarked five text-to-image **diffusion alignment** models (Diffusion-DPO, MaPO, etc.) with automated metrics and a GPT-4 preference study, where the **proposed DCPO-h model** achieved **+1.8 ImageReward** and a **66% human preference win rate** in visual appeal (TMLR 2025).
- Ran adversarial training on **7B–32B VLMs** with **DeepSpeed ZeRO-3**, **FlashAttention**, and multi-node GPU clusters to study robustness transfer; managed **SLURM orchestration** and performance profiling to improve training throughput and reduce OOM failures.
- Benchmarked three MLLMs against **six visual perturbations across VQAv2 and CLEVR**, revealing **20+% accuracy drops** and identifying **Gaussian filtering as most disruptive** for color and shape reasoning (CVPR 2024 Workshop).

Graduate Research Assistant (Advisor: Dr. Hasti Seifi)

Nov 2023 – Dec 2024

TEAL Lab, Arizona State University

Tempe, AZ

- Engineered a **unified Vision-Language Model framework** for **joint chart question answering and explanation generation**, achieving **+18.96% QA accuracy** and **+24.57 CIDEr gains** across supervised fine-tuned models.
- Designed structured data curation workflows including a **multi-stage pipeline** using **six SOTA VLMs** and **ROSCOE reasoning metrics** to produce **30K+ high-quality chart explanations**, validated by **245 AWS MTurk participants** for accuracy and logical consistency.
- Performed **SFT on VLM variants (7B-32B)** with DeepSpeed ZeRO-3 and FlashAttention, then second-stage **DPO and GRPO preference optimization** on ChartQA-X with vLLM evaluation; published as **"ChartQA-X"** (WACV 2026) with **+14.75% generalization gain on unseen chart datasets**.

Software Engineer Intern

Feb 2023 – May 2023

Bosch Global Software Technologies

Bengaluru, India

- Resolved **20+ production bugs** across distributed modules of a customer platform DeviceBridge, **improving uptime and stability**.
- Integrated **IronPython scripting** enabling users to write business logic for data ingested from **40+ device types** such as PLCs and sensors.
- Enabled **uniform data transformation and mapping** to MES, cloud, and database systems, streamlining industrial data flow.

PUBLICATIONS

- Hegde, S., et al. [ChartQA-X: Generating Explanations for Visual Chart Reasoning](#) - WACV 2026
- Saeidi, A.*, Luo, Y. L.*, Hegde, S., et al. [Dual Caption Preference Optimization for Diffusion Models](#) - TMLR 2025
- Hegde, S., et al. [Evaluating Multimodal Large Language Models Across Distribution Shifts and Augmentations](#) - CVPR 2024 Workshop
- Hegde, S., et al. [Making the V in Text-VQA Matter](#) - CVPR 2023 Workshop
- Hegde, S., et al. [Weakly Supervised Visual Question Answer Generation](#) - CVPR 2023 Workshop

RELEVANT PROJECTS

ShadowGuard: Real-Time PHI Detection - TreeHacks 2026 @Stanford, Healthcare Track Winner (1st Place, OpenEvidence)

- Built a **real-time ML inference system** intercepting live HTTPS traffic through a reverse-proxy **PHI detection pipeline** using a **local LLM** prompt-engineered to flag **healthcare-specific entities** (MRN, SSN, ICD-10 codes, medication names); scored exposure risk by entity type, frequency, and payload sensitivity, then **rewrote payloads before forwarding to external LLM APIs** (OpenAI, Anthropic, Google).
- Implemented **async event streaming** with FastAPI and WebSockets to push pipeline outputs to a **real-time compliance dashboard**, with an **automated voice alert system** using Elevenlabs that triggers phone calls using Vapi on large-scale PHI exposure events.

AutoScout: Intelligent Web Monitoring Agents

- Built an agentic system using Gemini, LangChain, and Playwright to **extract structured fields from dynamic web pages and detect content changes**, replacing manual website monitoring with a **real-time alerting pipeline**; deployed on a **serverless FastAPI + AWS Lambda backend** with caching and fallback extraction for template changes.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Java, JavaScript, Bash, HTML, CSS

ML & Modeling: PyTorch, TensorFlow, DeepSpeed, FlashAttention, vLLM, FSDP, SLURM, LangChain; LLMs, MLLMs, Diffusion Models, SFT, RLHF/DPO/GRPO, RAG, PEFT/LoRA

Cloud & Backend: AWS (S3, EC2, Lambda, SageMaker), Docker, CI/CD, FastAPI, WebSockets, REST

Databases: SQL (PostgreSQL, MySQL), NoSQL (MongoDB, DynamoDB), VectorDB (ElasticSearch, Pinecone)

EDUCATION

Arizona State University - Master of Science in Computer Science

Aug 2023 – Jul 2026

KLE Technological University - Bachelor of Engineering in Computer Science

Aug 2019 – Jul 2023